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Ethical Issues in the World of Cyber Security

Cybersecurity is a crucial field in today's world, as computers are an important part of society. When used correctly, it is one of the most ethical and helpful careers. However, this field is also extremely dangerous when the power falls into the wrong hands. Some of the biggest ethical issues that arise with cybersecurity are the mishandling of precious data, AI acting unethically, and potential insider threats. Fortunately, there are plenty of ways to avoid these issues. As Christians, we must ensure that these unethical issues do not occur, and we work to serve others and abide by the codes of ethics.

In the field of cybersecurity, each person is entrusted with extremely important and private information. One of the most prominent issues that people face is the possibility of leaking precious data or causing a data breach to occur. Cybersecurity professionals must practice confidentiality (Augusta University). The university says, "disclosing this data can have severe consequences, so cybersecurity professionals must never reveal confidential information, unless a significant public benefit exists for doing so" (AU). I am rather accident prone, so this is a potential issue for me. Of course, not all data breaches are intentional, but whether intentional or not, data breaches are one of the worst things that can happen in the field. It is imperative that professionals maintain confidentiality so people's precious information is not leaked, especially to the wrong people.

Using AI in cybersecurity raises ethical concerns as well. AI is a rapidly growing technology and is rather unavoidable at this stage. It can cause concerns regarding user safety because of its capacity to process large amounts of data (ISC2). Another problem AI causes is bias and fairness. According to ISC2, "In cybersecurity, a biased AI could result in profiling or unfairly targeting certain groups" (ISC2). Accountability is a large part of cybersecurity, and with AI, it is hard to determine who is accountable for a mistake that it causes (ISC2). The last problem is job displacement and economic issues. On top of the threat of job displacement, the dilemma "extends beyond the immediate concerns of cybersecurity professionals to broader societal implications, including economic impact and the need for retraining and reskilling" (ISC2). AI is not inherently bad, but if we do not take the time to make sure we use it ethically, it can cause more damage and ethical problems than help.

Of course, as with most things, there is always the concern of people using their abilities for malicious purposes rather than helping others. The main problem of this category is insider threats. According to the "Cybersecurity & Infrastructure Security Agency," an insider threat is "the potential for an insider to use their authorized access or understanding of an organization to harm that organization" (CISA). An insider is a person with authorized access or knowledge of the resources in an organization, like systems, networks, or equipment (CISA). There are two types of insider threats, unintentional threats and intentional threats. One unintentional threat includes negligence, which is exposing precious data because of carelessness and ignoring policies (CISA). The other one is accidental, which is a mistake that puts the organization at risk, like mistyping an email address or opening an attachment in a phishing email and giving the system a virus (CISA). An intentional threat is "actions taken to harm an organization for personal benefit or to act on a

personal grievance" (CISA). Insider threats, whether intentional or not, are extremely dangerous in the world of cybersecurity, and both are a serious ethical concern.

Data breaches are easily avoidable as long as one takes the right steps. They can be prevented in a multitude of ways. People can call out apps and designs that encourage users to share an unnecessary amount of personal information, mitigating risks, making sure others comply with security frameworks, and implementing countermeasures like risk assessments, firewalls, and continuous monitoring (AU). I can act ethically by ensuring that risks proper security measures are taken to avoid these potential breaches. I can also ensure that I am careful so that I do not accidentally cause a data breach myself.

AI is a very useful tool in the computer world, but AI is also quite flawed and unable to differentiate right from wrong. I feel like trying to just ignore AI will do no good, as you cannot truly avoid it in this field. These best ways to handle this problem is to mitigate bias, establish a clear accountability framework, continuous ethical training, and regular audits and assessments (ISC2). I do not typically use AI in my day-to-day life, so I feel that I can handle this issue well. I can also make sure to check the facts and make sure I am not using unethical AI. I can ensure that I maintain these standards and train as hard as I can to prevent AI from doing anything unintentionally unethical.

Insider threats are not as easy to prevent as you can never truly tell what someone is thinking or what their intentions are. However, organizations can take certain actions to ensure that insider threats happen as little as possible. CISA says that whenever an assessment of the organization reveals the potential of an insider threat, "the threat management team should recommend and coordinate approved measures to continuously monitor, manage, and mitigate the risk of harmful actions" (CISA). When I enter the field, the main way I can avoid this is by not

participating in such actions, and if I did hear anything about a potential threat, I can inform the authorities to help mitigate the threat. However, I feel I may struggle with the second part, as I tend to try to stay out of confrontation and issues that do not involve me. It will be important for me to push those feelings down though, as that I can serve others in an ethical way.

As Christians, it is important to follow the codes of ethics presented. Of course, the main code of ethics one should follow is the Bible, but programmers should also follow some computing codes of ethics, as they agree with what the Bible says. For example, one of the rules ACM presents is “be honest and trustworthy” (AMC, 12-14). Ephesians 4:25 says, “Therefore each of you must put off falsehoods and speak truthfully to your neighbor, for we are all members of one body” (NIV, Eph. 4:25). The book of Proverbs explains that the Lord detests liars, but delights in trustworthy people (NIV, Prov. 12:22). One final verse is “Do your best to present yourself to God as one approved, a worker who does not need to be ashamed and who correctly handles the word of truth,” found in 2 Timothy 2:25 (NIV, 2 Tim. 2:25). Another code of ethics, the IEEE, also agrees with the Bible. The first rule is “to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment” (IEEE, 3). There are several verses that talk about serving others, such as Matthew 5:16. It says, “In the same way, let your light shine before others, that they may see your good deeds and glorify your Father in heaven” (NIV, Matt. 5:16). The book of Proverbs reads that those who are generous will prosper, and those who refresh others will be refreshed (NIV, Prov. 11:25). It is important to follow the computer code of ethics we are given, but we must also ensure to follow the Bible, as it is the first code of ethics anyone should follow.

The world of cybersecurity offers a myriad of ways to serve others, but it also has many ways to cause serious damage and harm to others. We must ensure that unethical practices do not occur. Some of the practices to avoid are data breaches, using unethical AI, and insider threats. As a Christian and protector of precious information, I must make sure that I follow what the codes of ethics and Bible say and be prepared for any situation that could be unethical.

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